Administer Oxygen
To maintain Sp02 >94%

Rule out foreign body obstruction as cause

- Keep child calm
- Avoid unnecessary distress
- Allow for position of comfort

Mild
(0-2 points)

Moderate
(3-7 points)

Severe
(8+ points)

Consider ALS Back-Up

Consider ALS Back-Up

EMT

Consider Nebulized Racemic Epinephrine (2.25%)
Dose: 0.5ml
May be repeated after 20 minutes if required.
Max 3 doses

Nebulized Racemic Epinephrine (2.25%)
Dose 0.5ml
May be repeated after 20 minutes if required.
Max 3 doses

EMT

Avoid airway stimulation as it may increase airway swelling and obstruction

Reassess

Assessment of Croup Severity[6]						
	0	1	2	3	4	5
Inspiratory Stridor	Not Present	When Agitated / Active	At Rest			
Intercostal Recession	No recession	Mild recession	Moderate recession	Severe Recession		
Air Entry	Normal	Mildly decreased	Severely decreased			
Cyanosis	None				With Agitation / Activity	At rest
Level of Consciousness	Normal					Altered

The total score is then categorized as follows:
- **Mild** croup (0-2 points)
- **Moderate** croup (3-7 points)
- **Severe** croup (8-11 points)
- **Impending Respiratory Failure / Life Threatening** (≥12 points)

Patient Care Goals

- Relieve upper airway obstruction.
- Reduce patient distress while maintaining oxygenation.
- Avoid unnecessary agitation that may worsen obstruction.
- Safely transport to definitive care if required.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with Croup Symptoms, including;
  - Typically children aged 6 months–6 years.
  - Barking, "seal-like" cough.
  - Inspiratory stridor at rest or with agitation.
  - Hoarseness, respiratory distress.
  - Fever (usually low-grade).

Exclusion Criteria

- Signs of epiglottitis (drooling, dysphagia, sitting forward, very high fever).
- Foreign body aspiration (sudden onset, asymmetry).
- Anaphylaxis.
- Severe bacterial tracheitis.
- Asthma (or other causes of bronchospasm)

Key Patient Assessment Points

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey's as per Universal Care CPG.
   - **Initial/General Impression**
     - Perform from a distance at first, to avoid agitating the child.
     - Use the Pediatric Assessment Triangle (PAT):
       - Appearance: alertness, tone, interaction, look for lethargy or agitation.
       - Work of Breathing: stridor, retractions, nasal flaring, tracheal tug, accessory muscle use.
       - Circulation to Skin: pallor, cyanosis, mottling , signs of hypoxia or poor perfusion.
   - **Airway Assessment**
     - Listen for barking "seal-like" cough, inspiratory stridor, and hoarseness.
     - Assess for obstructed airway signs:
       - Drooling (suggests epiglottitis rather than croup).
       - Difficulty swallowing.
       - Upper airway noises at rest versus exertion.
     - Assess if stridor is at rest (more concerning) or only on exertion/crying.
   - **Breathing Assessment**
     - Rate: tachypnea or slow (impending failure).
     - Effort:
       - Mild: minimal retractions, no stridor at rest.
       - Moderate: visible retractions (suprasternal, intercostal, subcostal), stridor at rest.
       - Severe: marked retractions, nasal flaring, fatigue, "silent chest" (air movement reduced).
     - Oxygen saturation (SpO_2): aim for ≥94%; lower indicates hypoxia.
   - **Circulation Assessment**
     - Heart rate: elevated in distress, bradycardia in severe hypoxia.
     - Capillary refill time: delayed if poor perfusion.
     - Skin color: cyanosis (late, ominous), mottling, pale.
   - **Mental Status Assessment**
     - Observe for fatigue, reduced level of consciousness, or confusion (signals exhaustion and impending respiratory failure).

- Irritability or agitation may worsen obstruction, but lethargy is more concerning.
- **History Taking** (SAMPLE & OPQRST)
  - S (Symptoms): barking cough, stridor at rest or with activity, hoarseness.
  - A (Allergies): prior reactions to medications.
  - M (Medications): any prior steroids given at home.
  - P (Past medical history): previous croup episodes, airway abnormalities, asthma, hospitalizations, immunization history.
  - L (Last oral intake): dehydration risk.
  - E (Events): onset, progression, triggers (often viral, seasonal).
  - Fever: usually mild (<39°C); higher fevers or drooling suggest alternate diagnosis.

2. **Severity Assessment**
   - Assess Croup Severity

Assessment of Croup Severity[6]						
	0	**1**	**2**	**3**	**4**	**5**
Inspiratory Stridor	Not Present	When Agitated / Active	At Rest			
Intercostal Recession	No recession	Mild recession	Moderate recession	Severe Recession		
Air Entry	Normal	Mildly decreased	Severely decreased			
Cyanosis	None				With Agitation / Activity	At rest
Level of Consciousness	Normal					Altered
The total score is then categorized as follows: • **Mild** croup (0-2 points) • **Moderate** croup (3-7 points) • **Severe** croup (8-11 points) • **Impending Respiratory Failure / Life Threatening** (≥12 points)						

3. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority (Serious & Life-threatening):
       - Severe respiratory distress with hypoxia, lethargy, cyanosis, or impending respiratory failure.
       - Persistent stridor at rest, biphasic stridor, silent chest, poor air entry despite intervention.
     - Yellow Priority (Serious but not immediately life-threatening):
       - Moderate distress with stridor at rest, increased work of breathing, but stable oxygenation and mental status.
       - Responds to initial intervention but still symptomatic.
     - Green Priority (Mild & Stable):
       - Mild symptoms with stridor only when agitated, no hypoxia, no signs of exhaustion, stable mental status and perfusion.

Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for:
     - Resting or bi-phasic stridor.
1. **Keep the child calm and minimize distress**
   - Maintain a quiet environment, allowing the child to remain with caregivers in a position of comfort, and avoiding unnecessary interventions.
   - Avoid procedures or handling that could increase agitation, which may worsen airway obstruction.
2. **If the child exhibits moderate or severe respiratory distress:**
   - Croup Severity Score >3 points.
   - **Consider nebulized racemic epinephrine** (2.25% solution):
     - Dose: 0.5 mL diluted in 2.5–3 mL of normal saline, administered via nebulizer over ~15 minutes.
     - May be repeated every 20–30 minutes as needed, observing for improvement and recurrence of symptoms.
     - Maximum 3 doses without consultation with Tele-EMS.
   - If racemic epinephrine is unavailable, standard L-epinephrine (1:1,000) at 0.5 mL/kg (max 5 mL) nebulized may also be used.
   - Any patient who has received nebulized epinephrine must be transported to hospital
3. **Provide supplemental oxygen if SpO2 <94%**
   - Target SpO2 ≥94% and <98%, using the least intrusive method tolerated.
   - If SpO_2 falls below 92%, escalate interventions and consider urgent hospital notification.
4. **Prioritize Transport**
   - All children receiving racemic epinephrine or who remain in moderate or severe distress should be transported promptly to hospital.
5. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Airway:
       - Any audible change in stridor intensity or loss of audible air movement (potential impending failure).
     - Breathing:
       - Work of breathing: stridor at rest, retractions, nasal flaring, tracheal tug, silent chest.
       - Oxygenation: pulse oximetry (SpO_2) — maintain ≥94%, watch for any downward trend.
       - Respiratory rate and pattern: monitor for tachypnea, slowing (which may indicate fatigue), irregular breathing.
     - Circulation:
       - Heart rate: persistent tachycardia may indicate ongoing distress; bradycardia is a late and serious sign.
     - Response to treatment:
       - Response to racemic epinephrine: assess if improvement plateaus or symptoms recur, which may necessitate repeat dosing.
   - Be prepared to intervene if the child's condition deteriorates, with equipment and medications accessible and ready.
   - Ensure clear communication with the receiving facility regarding interventions given, response, and current clinical status.

Patient Safety Considerations

- Avoid unnecessary interventions that increase agitation.
- Ensure airway equipment is ready but avoid invasive airway procedures unless absolutely necessary.
- Be prepared for rapid deterioration in severe cases.

- Monitor during and after adrenaline, as symptoms may recur once it wears off.
- Any patient who has been administered nebulized adrenaline must be transferred to hospital.

Documentation

- Complete a comprehensive electronic Patient Care Record (ePCR), capturing the following elements to support continuity of care. Including:
    - Identification — name, age, gender.
    - Mechanism/Medical complaint — croup: barking cough, stridor.
    - Injuries/Information — fever, severity, any prior episodes or treatments.
    - Signs — vital signs (RR, HR, SpO_2), degree of respiratory distress.
    - Treatment — oxygen, steroids, nebulized adrenaline.
    - Allergies — especially to steroids or adrenaline.
    - Medications — any prior steroids, antipyretics.
    - Background — past medical history (e.g., asthma, previous croup, airway anomalies).
    - Other — caregiver observations, social circumstances, decision to transport or not.

Key Performance Indicators

1. Administer racemic epinephrine for moderate to severe cases.
2. Administer oxygen to patients with an SpO_2 <94%

Clinical Insights & Additional Guidance

- Croup is usually a self-limiting viral illness, but in some cases airway obstruction can progress rapidly.
- Symptoms often worsen at night and may recur within hours of initial improvement, particularly after the effects of racemic epinephrine wear off.
- Mild cases can often be managed at home with caregiver education, provided the child maintains adequate oxygenation and no stridor at rest.
- Moderate to severe cases require hospital evaluation and monitoring, as repeated doses of racemic epinephrine or further interventions may be necessary.
- Request ALS back-up early if moderate or severe distress, impending failure, or deterioration expected, as per Universal Care CPG.
- Consult Tele-EMS when in doubt about management decisions or disposition.
- Differentiate croup from other serious upper airway conditions (epiglottitis, bacterial tracheitis, foreign body aspiration) which may present similarly but require different management.
- Ensure caregivers understand the temporary nature of racemic epinephrine's effect, the potential for symptom recurrence, and when to seek further help.
- Avoid humidified air and unnecessary interventions, as these have not been shown to improve outcomes and may increase distress.
- Always prepare for possible airway compromise and have equipment readily available. Avoid invasive airway management unless absolutely necessary.

References

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